

NASA'S TBIRD MISSION

Protocol Design & Error Correction

Asymmetric Routing and ARQ in 200-Gbps Space-to-Ground Optical Networks

Karanveer Singh | CIS 419: Data Communications and Computer Networking

CHALLENGE

The Problem with TCP in Space

- ↔ **Symmetric Reliance:** Traditional TCP relies on symmetric, in-band acknowledgments (ACKs).
- 🔔 **False Congestion:** TCP assumes packet loss is caused by network congestion.
- ⚡ **Physical Disruptions:** In space-to-ground optical links, packet loss is caused by physical layer disruptions like weather and atmospheric fading.
- 🚫 **The Result:** TCP dramatically throttles speed, failing to utilize the 200 Gbps pipeline.

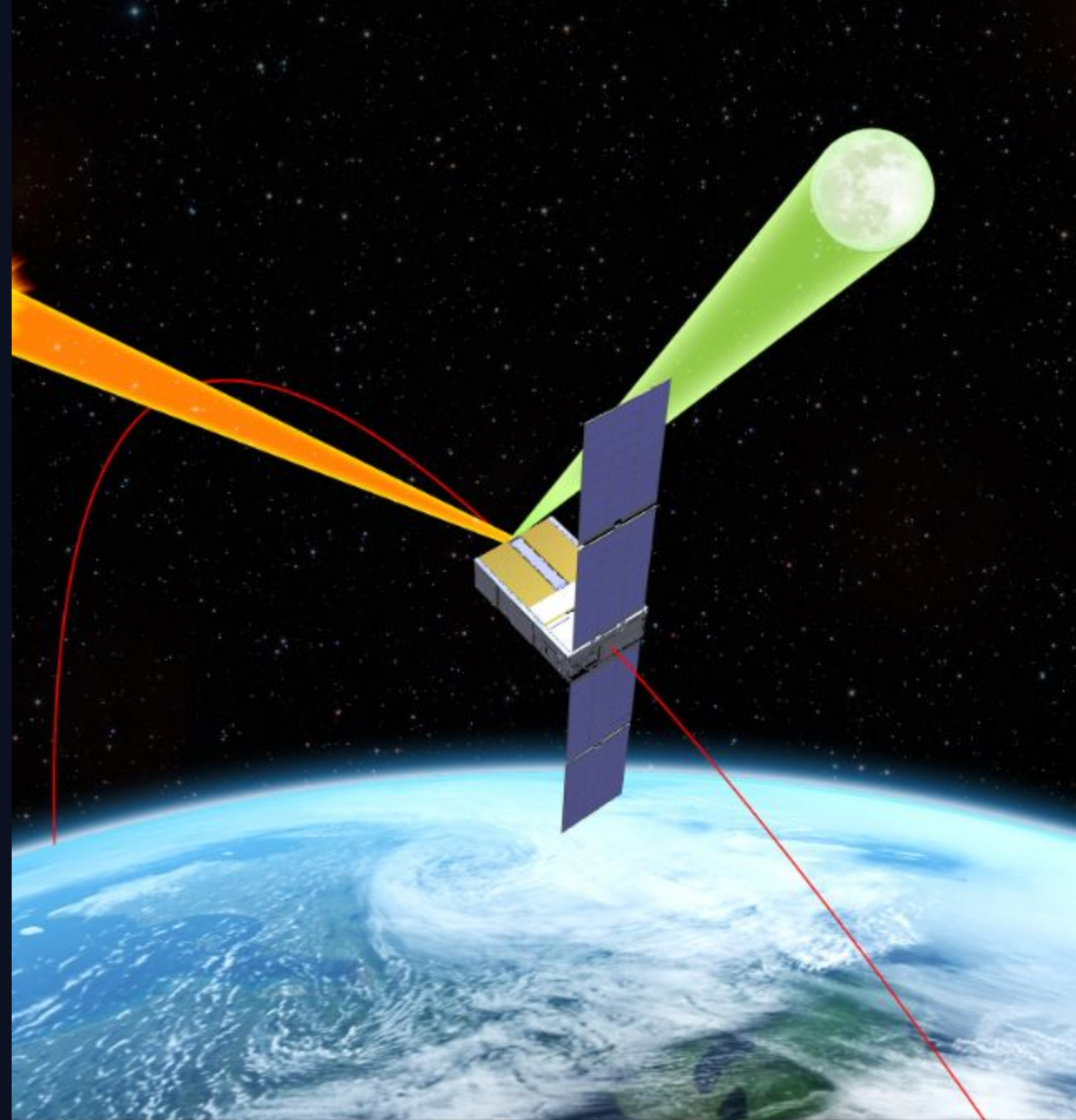


OVERVIEW

TBIRD Mission

TeraByte InfraRed Delivery

-  **Joint Venture:** Collaboration between NASA and MIT Lincoln Laboratory.
-  **Primary Goal:** Demonstrate extreme high-bandwidth data transfers from Low Earth Orbit (LEO) aboard 6U PTD-3 CubeSat.
-  **Achievement:** Record-breaking 200 Gbps optical downlink.
-  **Constraint:** Passes over the ground station only last about 5 minutes.



ARCHITECTURE

The Multi-Medium Solution

This creates a highly asymmetric routing environment where data and ACKs travel over entirely different physical mediums.

Downlink (Data Payload)

200 Gbps Optical Laser

Uplink (Acknowledgments)

2 kbps Radio Frequency (RF) link



IMPLEMENTATION

Bespoke ARQ Protocol



Replaces TCP

Completely replaces standard TCP's congestion-control logic to maintain optimal transmission speeds through space.



Massive Buffering

Implements specialized high-capacity frame buffering at the ground station receiver.



Selective Repeat

The low-bandwidth RF uplink requests retransmission only for specific missing frames rather than halting the stream.

COMPARISON

Algorithmic Adaptation

Feature	Standard TCP	TBIRD Custom ARQ
ACK Path	In-band (Symmetric)	Out-of-band (Asymmetric / RF)
Loss Reaction	Reduce Window Size	Maintain speed, re-queue missing frames
Primary Bottleneck	Congestion / Buffer Bloat	Atmospheric Fading / Weather

SUMMARY

Conclusion & Future Implications

Standard protocols are tightly coupled to terrestrial physical constraints.

Extreme bandwidth (200 Gbps) requires decoupling the data path from the control path.

TBIRD's architecture paves the way for the next generation of interplanetary data communications.

References

Schieler, C. M., et al. (2023)	"Recent On-Orbit Results and ARQ Performance Analysis for the TBIRD 200-Gbps Mission." NASA Technical Reports Server.
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